2 Fig. 2.1 shows water stored in a reservoir behind a hydroelectric dam.

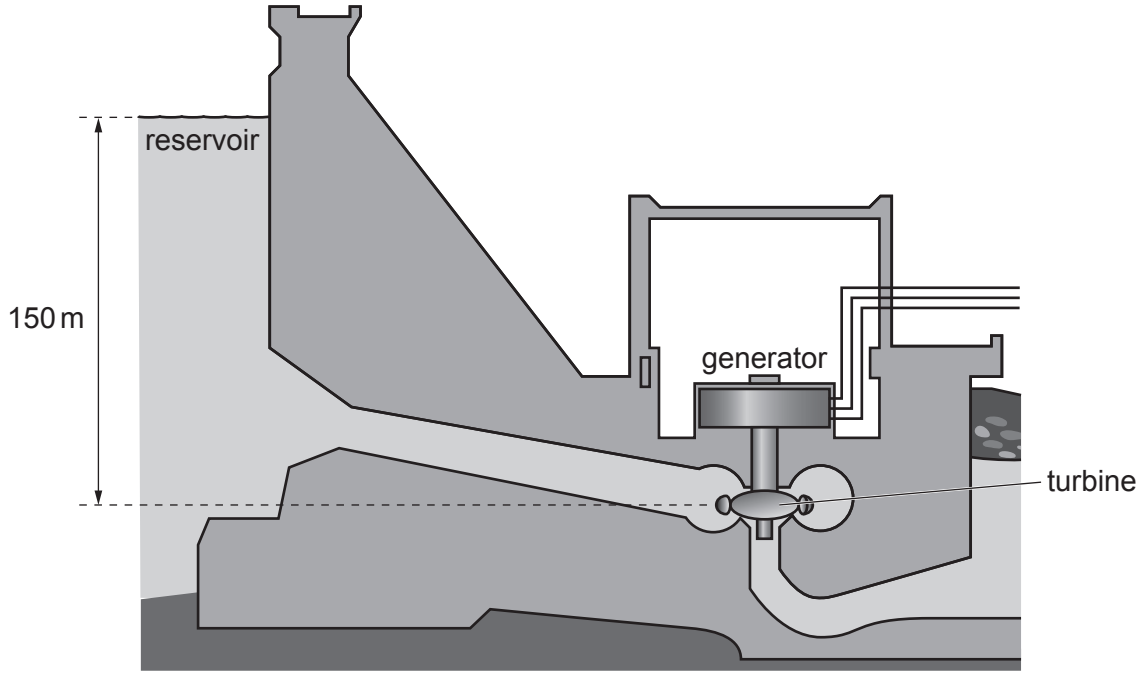


Fig. 2.1 (not to scale)

(a) State the form of the energy stored in the water in the reservoir that is used to generate electricity.

..... [1]

(b) The turbine is 150 m below the level of the water in the reservoir.

Atmospheric pressure is 1.0×10^5 Pa. The density of water is 1000 kg / m^3.

(i) Calculate the total pressure in the water at the turbine.

pressure = ... [3]

(ii) The turbine has a cross-sectional area of $3.5\,m^2$.

Calculate the force exerted on the turbine by the water.

force = ... [2]

(c) The water flows to the turbine through a pipe of constant cross-sectional area.

Explain why the kinetic energy of the water in the pipe remains constant as it flows through the pipe.

.....

.....

..... [2]

[Total: 8]